

Developer and Publisher



Image sources: <https://tiermaker.com/categories/video-games/game-developers-definitive-edition-54837>



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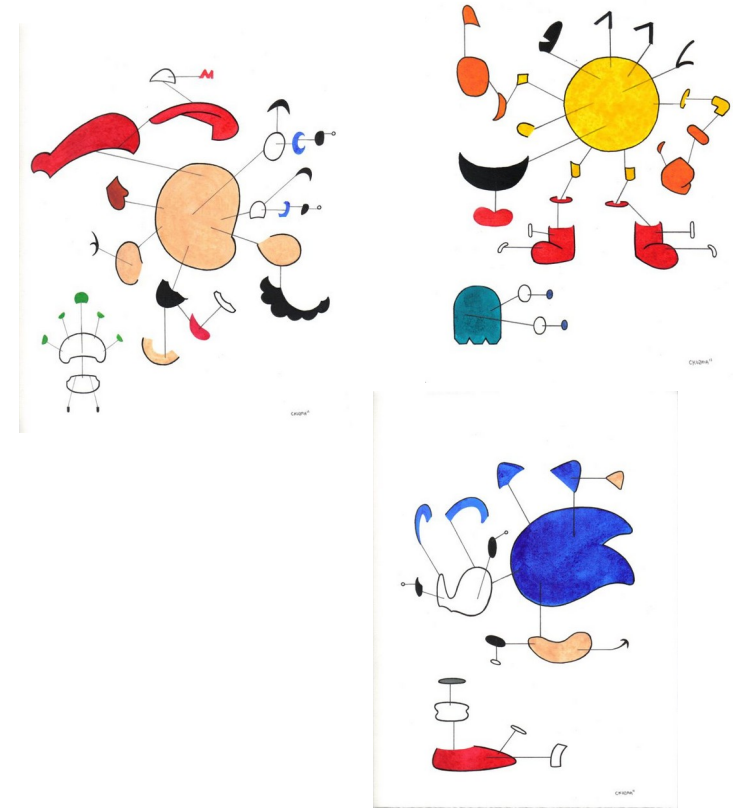


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Components of a game



- What does a game consist of?
- What needs to be created and how?
- Source Code
 - Game Engine, Game Logic, Tools, ...
- Graphics Assets
 - 3D Models, Textures, Videos, Effects, ...
- Sound Assets
 - SFX, Music, Speech, Atmosphere, ...
- Game Design
 - Story, Characters, Level Design, Dialogues, ...



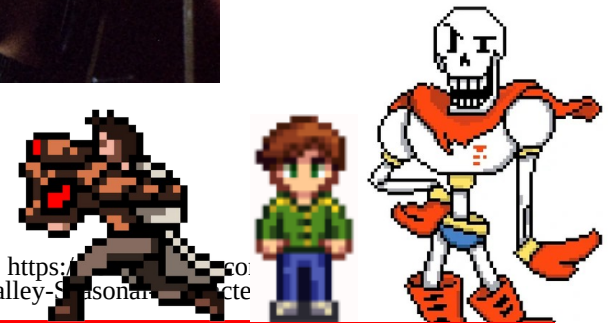
The good old times...



- One-Man-Team
 - The original Prince of Persia was made in 1989 by one person: Jordan Mechner
 - Programming, Design, Graphics, Animation, Sound, Music, ... everything!



Image sources: <http://mygaming.co.za/news/news/36296-jordan-mechner-releases-original-prince-of-persia-source-code.html>, <https://www.epicgames.com/store/en-US/product/axiom-verge/home>, <https://www.deviantart.com/rjenyawd/art/Stardew-Valley-Seasonal-Character-Concepts>



The present day



- Games today are giant projects with multi-million dollar budgets (much like movies).
 - *Star Citizen*: 750+ million US\$ combined (est).
 - *Cyberpunk 2077*: 174 million US\$ development + 142 million US\$ marketing.
 - *Grand Theft Auto V*: ~137 million US\$ development + ~110 million US\$ marketing (unofficial figure).
- Creation of a modern video game takes several years of development time.
- Development teams consist of several highly specialized creative people all working together to create entertainment.
- Game teams are usually composed of the following disciplines:
 - Engineer
 - Artist
 - Designer
 - Quality Assurance
 - Producer
 - Support staff

Engineers (Programmers)



- Programmers design and implement the software that makes the game.
- Runtime programmers
 - Work on the engine and the game itself
- Tool programmers
 - Work on the offline-tools that allow the rest of the development team to work effectively
- Specialists
 - Rendering, artificial intelligence, audio, collision and physics, shader effects, gameplay, user interfaces, input, scripting, etc.
- Some engineers are generalists
 - Jump around and tackle whatever problem might arise



Image sources: <http://www.clipartpanda.com/categories/programming-20clipart>

Artists (1/3)



- Content is king!
 - Artists produce all of the visual and audio content in the game.
 - The quality of their work can make or break a game!
- Concept artists
 - Produce sketches and paintings of characters, worlds and enemies
 - Provide the team a vision of what the final game will look like
- 3D modelers
 - Produce the three-dimensional geometry for everything in the virtual game world
 - Foreground modelers
 - Characters, vehicles and other objects that populate the game world
 - Background modelers
 - Static background elements like buildings, terrain and bridges



Image sources: <http://worldartisme.com/clip-art-was-for-clipart.html>

Artists (2/3)



- Texture artists
 - Create two-dimensional images (textures) which are applied to the models in order to provide detail and realism
- Lighting artists
 - Lay out all of the light sources in game world (static and dynamic)
- Animators
 - Imbue the characters and objects with motion
 - They quite literally perform as actors (like in CG film productions)
- Visual effect artists
 - Create 2D and 3D special-effects like fire, smoke and explosions.
- Technical artists
 - Perform additional (invisible) tasks like rigging models and optimizing various content for performance



Artists (3/3)



- User interface artists
 - Create icons and all elements of the games interface
- Motion capture actors
 - Perform motions which are recorded in a mocap-studio
- Voice actors
 - Provide voices for characters
- Sound designers
 - Produce and mix sound effects
- Music composers
 - Compose the original score of the game



Game Designers



- Design the interactive proportion of the game's experience, also known as gameplay
 - Level designer, system designer, scripter, combat designer, creative director, ...
- Need a high degree of strength in communication skills
- Work on different levels of detail
 - Macro level
 - Story arc, overall sequence of chapters or levels, high-level goals or objectives of the player
 - Some game teams employ dedicated writers who construct the story arc or write dialog lines
 - Individual levels or geographical areas of the game world
 - Laying out static level geometry (“grey boxing”), defining enemy spawn points, placing supplies, designing puzzle elements, designing quests, ...
 - Technical level
 - Programming scripts, designing AI, balancing values and probabilities



Image sources: <https://openclipart.org/tags/board%20games>

Game Designers



- Design the interactive properties of the game's experience, also known as gameplay
 - Level design
- Need a high level of creativity
- Work on different scales
 - Macro level
 - Story
 - Sound
 - Individual level
 - Layout design
 - Technical
 - Programming



the player
log lines

ing supplies,

Image sources: <https://openclipart.org/tags/board%20games>

Quality Assurance (Testers)



- Play the game constantly in order to find bugs and flaws
 - Technical problems (bugs, crashes, HW specific problems, ...)
 - Design problems (difficulty, accessibility, ...)
- A good tester has patience, persistence and great communication skills to report back any problems
 - Other team members must be able to rely on the created reports
 - The tester has to re-play the same parts over and over again
- Quality assurance is crucial to the successful completion of a game
 - High standards of quality
 - Reliability and stability
- Testing is a great gateway job for newcomers
- Testers prevent games from sucking!



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Image sources: <https://www.illustrationsof.com/439944-royalty-free-gamer-clipart-illustration>

Producers



- Manage the schedule and human resources
 - Milestone plan (external)
 - Task plan (usually internal)
- Liaison between the development team and the business unit
 - Defend interests of both parties
- Take care of business duties
 - Development contracts
 - Contact with external partners
- Sometimes have game design responsibilities
 - Have to represent the product in front of the client
- Always appear on top of the credits screen... ;-)

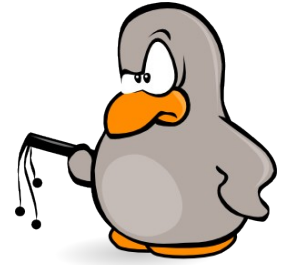


Image sources: <https://pixabay.com/en/penguin-linux-bird-manager-work-24598/>

Support staff



- Do not directly construct the game
- Studio executive manager
- Marketing department
 - Or a team that liaises with an external marketing group
- Administrative staff
- IT department
 - Purchase, install and configure HW and SW
 - Provide technical support
 - Engineers the backend e.g. Server infrastructure



Image sources: <http://www.clipartpanda.com/categories/provider-clipart>

Development studios (1/2)



Naughty Dog Inc, Santa Monica, California

Image sources: <https://www.gamedesigning.org/game-development-studios/>

Development studios (2/2)



- A video game developer is a software developer (a business or an individual) that creates video games.
 - Often specialized in certain types of games.
 - Some focus on porting or translation of games.
- Quality of life
 - Video game development is usually performed in an extremely casual business environment with T-shirts and sandals being common work attire.
 - However, the industry is also known to require generally high working hours of its employees, sometimes at a level seen as unsustainable and destructive. Employee burnout is not uncommon.
- Crunch time
 - The point at which the team is believed to be failing to achieve the milestones needed to launch the game on time.
 - Controversially, most game development engineers and artists are not paid for their overtime when crunching - ostensibly because they are salaried employees.

Types of developers (1/3)



<https://www.youtube.com/watch?v=zxR0sUT3egC>

Types of developers (2/3)



- First-Party developers
 - Game studios owned directly by a console manufacturer.
 - May either use the name of the company itself, or have a specific division name.
- Second-Party developers
 - Non-owned third-party developer that specializes in development for a particular platform.
 - May have exclusive publishing agreements or other business relationships with a particular manufacturer while maintaining independence.
 - May be wholly owned and funded by their parent company, but these are all considered subsidiaries.



GUERRILLA



GAME FREAK



Image sources: <https://en.wikipedia.org/wiki/...> Nintendo, Microsoft_Studios, Naughty_Dog, Valve_Corporation, Guerrilla_Games, Ready_at_Dawn, Game_Freak, Insomniac_Games

Types of developers (3/3)



- Third-Party developers
 - Are usually called upon by a publisher to develop a title for one or more systems.
 - Publisher and developer have a great deal of say as to the design and content of the game. But... the one with the money usually has the final say...
 - A volatile business! A single canceled game can be lethal for a small developer.
- Independent developers
 - Small software developers that are not owned by or beholden to a single publisher.
 - Digital distribution of inexpensive games makes it possible to get wide distribution for their games.



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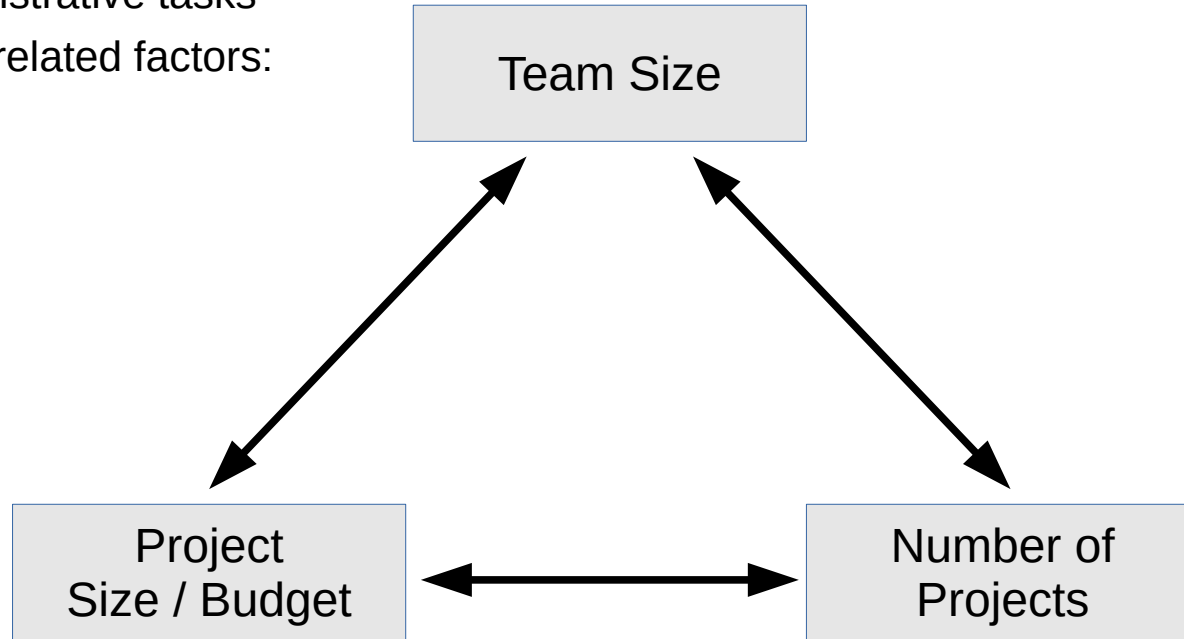


Image sources: <http://www.independent-arts-software.de/>, <http://www.redhookgames.com/>, <https://ska-studios.com/>, <https://lab132.com/>, https://en.wikipedia.org/wiki/..._Deck13, [Telltale_Games](https://en.wikipedia.org/wiki/..._Telltale_Games), [CD_Projekt](https://en.wikipedia.org/wiki/..._CD_Projekt), [Moon_Studios](https://en.wikipedia.org/wiki/..._Moon_Studios), [Yacht_Club_Games](https://en.wikipedia.org/wiki/..._Yacht_Club_Games), [Night_School_Studio](https://en.wikipedia.org/wiki/..._Night_School_Studio)

Team structure



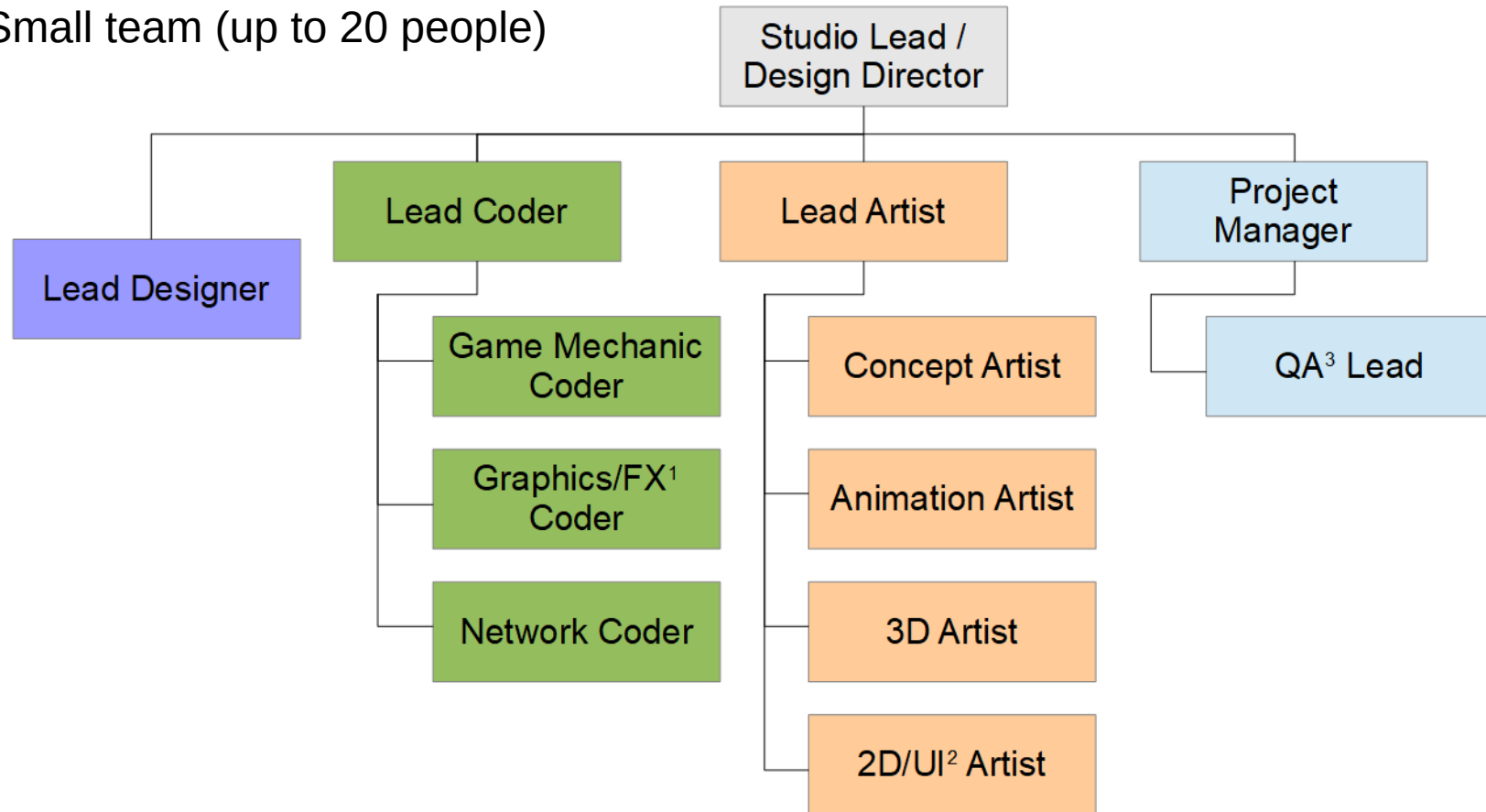
- How can a team of so many creative people be organized?
 - Communication overhead needs to be minimized
 - Team members don't always speak the same language...
 - Crucial administrative tasks
 - Several inter-related factors:



Team organization (1/4)



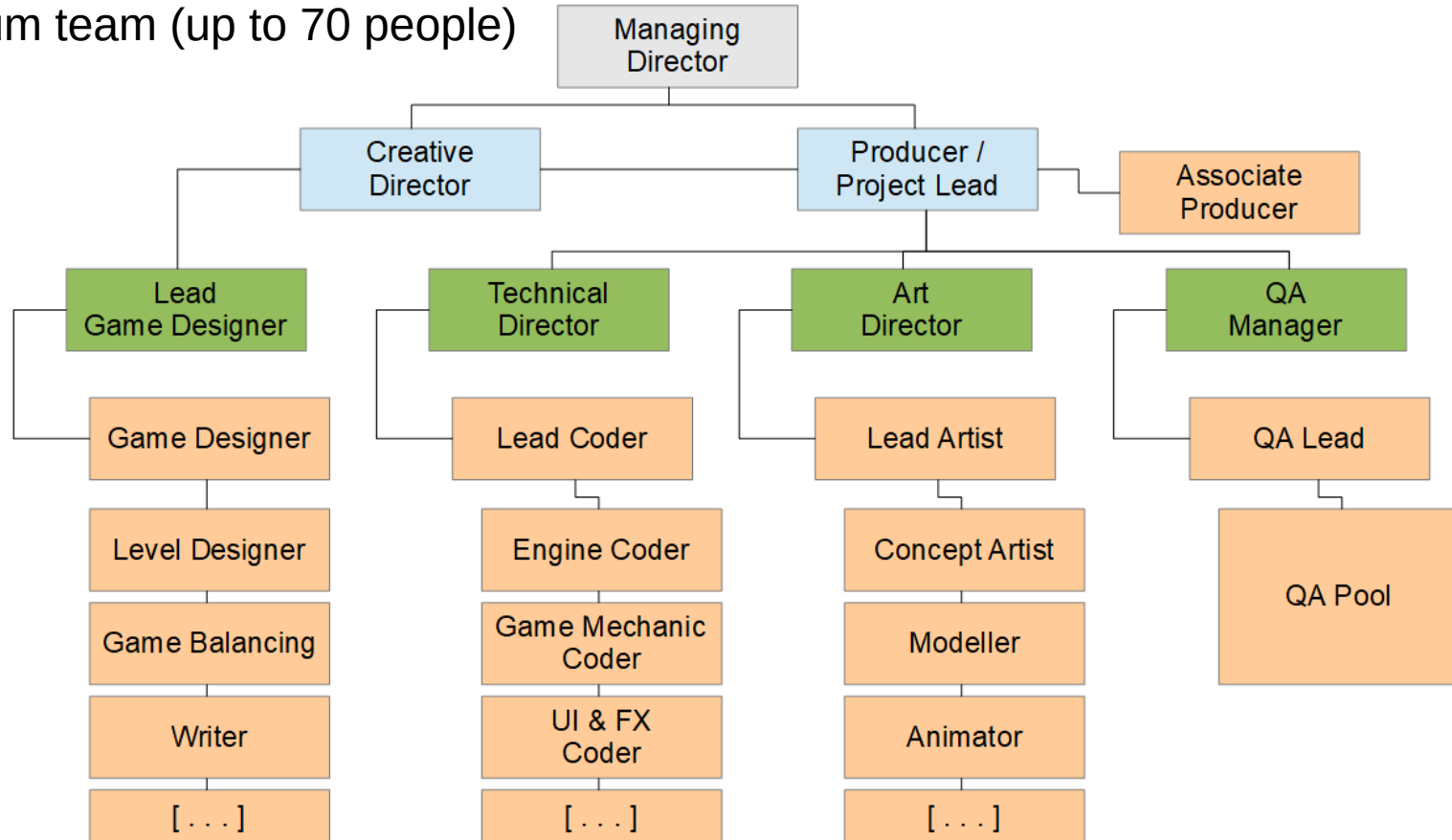
- Small team (up to 20 people)



Team organization (2/4)



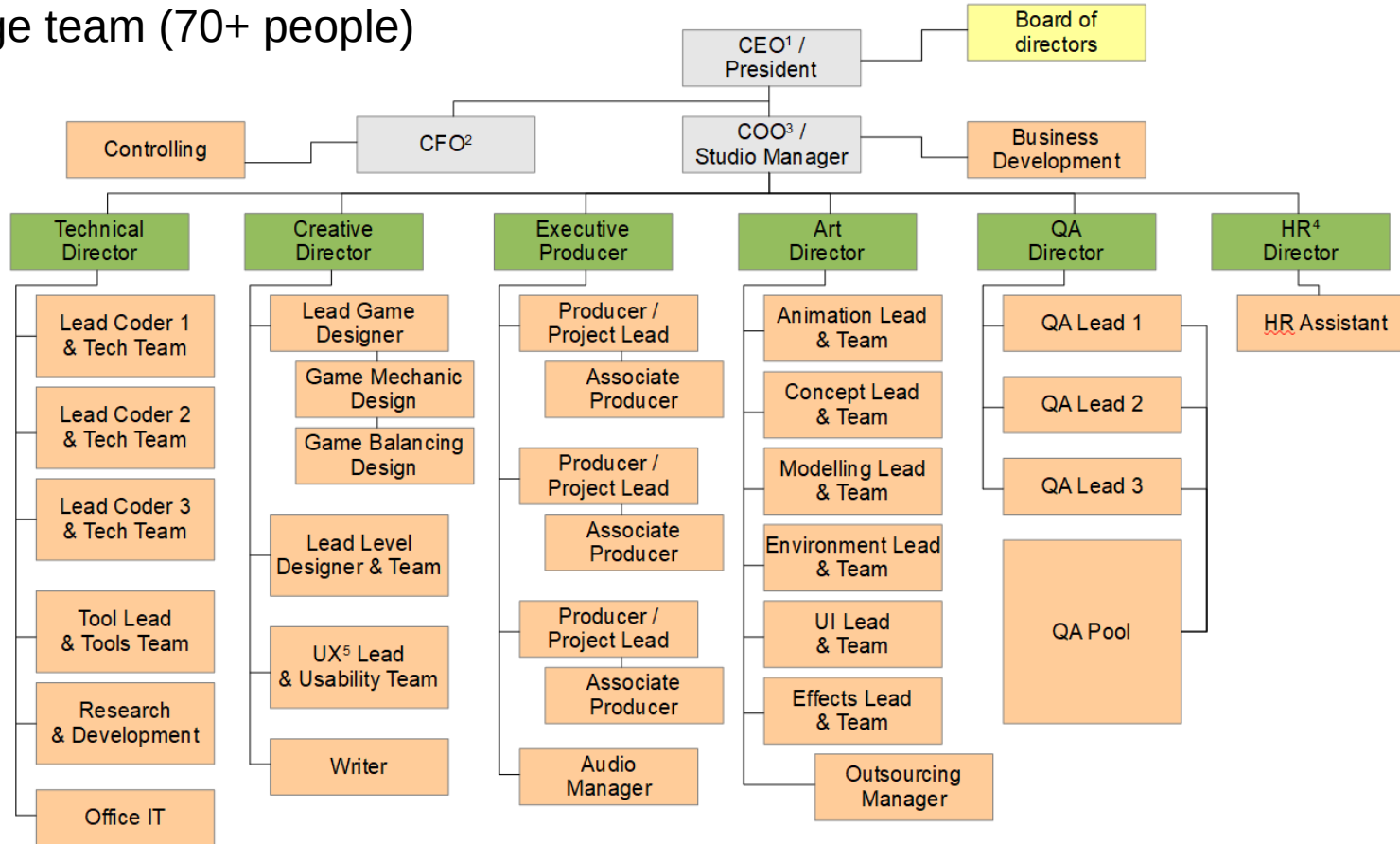
- Medium team (up to 70 people)



Team organization (3/4)



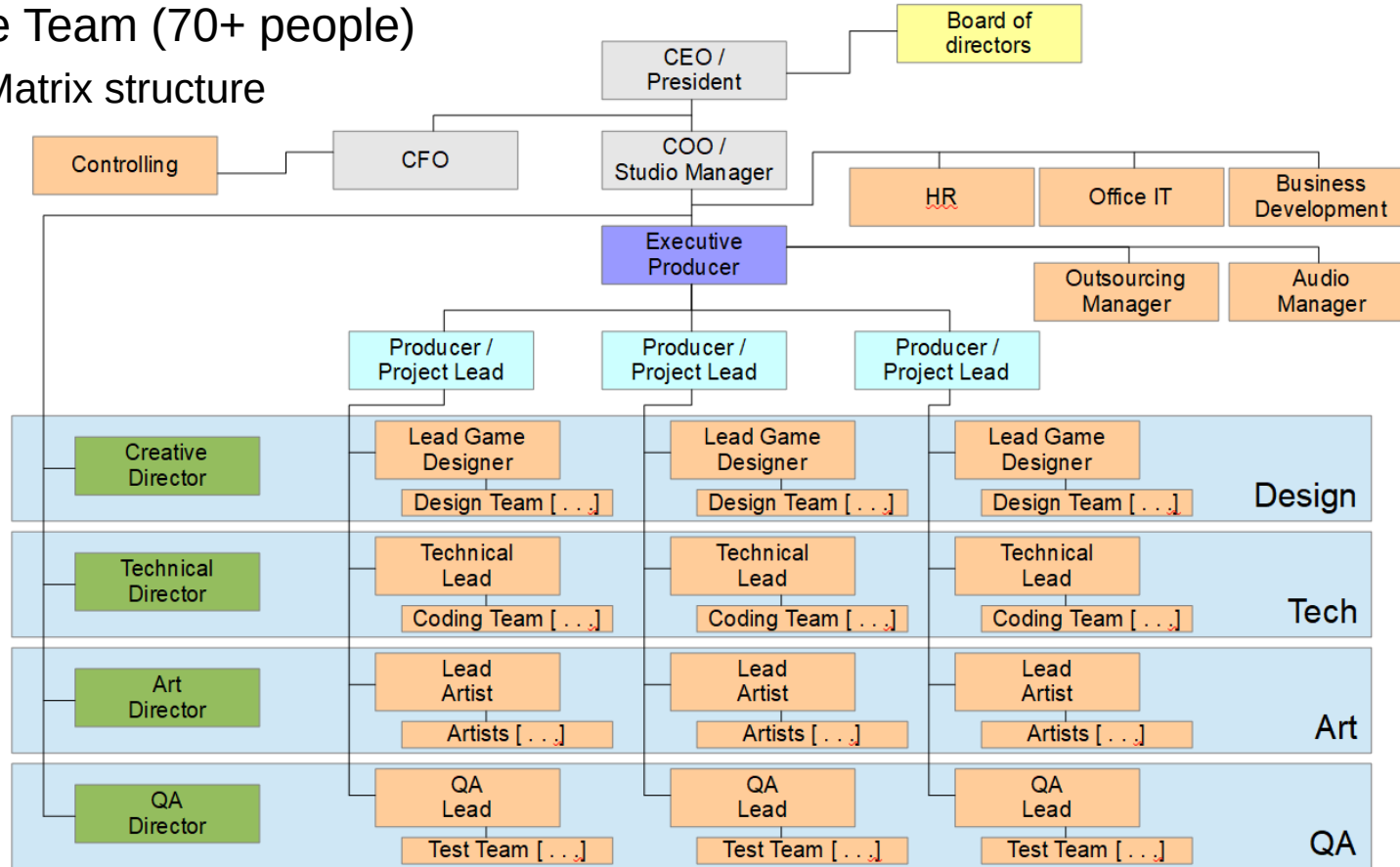
- Large team (70+ people)



Team organization (4/4)



- Large Team (70+ people)
 - Matrix structure



Publishers (1/2)



Image sources: https://www.reddit.com/r/gaming/comments/nkrvh9/who_is_your_favourite_publisherstudios_of_all_time/?rdt=54381

Publishers (2/2)



- A video game publisher is a company that publishes video games that they have either developed internally (second-party) or have had developed by a video game developer.
 - Responsible for their product's manufacturing and marketing, including market research and all aspects of advertising
 - They usually **finance the development**
 - Large publishers also distribute the games they publish, while some smaller publishers instead hire distribution companies
 - Also deciding on and paying for any license that the game may utilize
 - Paying for localization; layout, printing, and possibly the writing of the user manual; and the creation of graphic design elements such as the box design
 - Publishers usually try to manage **development risk** with a staff of producers or project managers to monitor the progress of the developer, critique ongoing development, and assist as necessary

Development contracts (1/3)

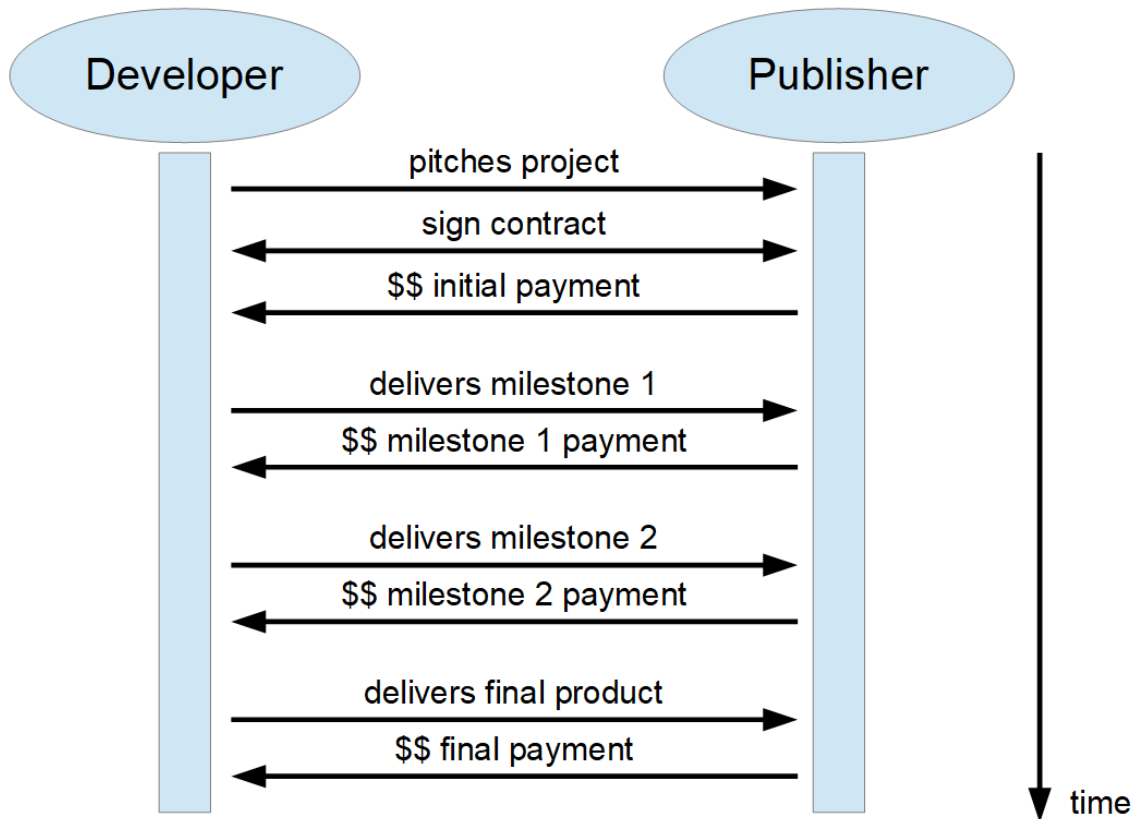


- Contracts specify a list of milestones intended to be delivered
 - The publisher is able to verify that work is progressing quickly enough to meet the projected deadline
 - The publisher can give direction to the developer if the game is turning out other than as expected in some way
 - This is almost always the case...
 - When each milestone is completed and accepted, the publisher pays the developer an advance on royalties
 - Acceptance criteria is defined in advance
 - Evaluated by the publisher
 - Payments may be reduced if the criteria isn't met completely (percentage)
- Possible profit-sharing if the game is successful
 - Examples:
 - 5% of the profit per unit after 10.000 units have been sold
 - Participation of 10% of every in-game-purchase

Development contracts (2/3)



- Milestone contract timeline example



Development contracts (3/3)



- Possible questions regarding development contracts:
 - What if a milestones' acceptance criteria is not met by 100%?
 - What if the developer doesn't deliver a milestone in time?
 - What if the publisher refuses or misses to pay?
 - What if the publisher cancels the game in the middle of production?
 - What about deliverables from the publisher?
 - What about contractual penalties?
 - What if the publisher makes billions of dollars selling the game?
 - Who is in trouble if the game turns out to be a financial flop?
 - Can one developer have multiple of these contracts at once?
 - What role do in-game-purchases play for the developer?
 - Who owns the utilization rights?
 - What happens if either publisher or developer become bankrupt?
 - ... can you think of even more?

What about self-publishing?



- Nowadays, it is “easy” for Indies to self-publish their games
- Benefits
 - You get 100% of the earnings (minus platform fees)
 - You get to keep your intellectual property
 - Flexible deadlines and release schedules
 - 100% creative freedom
- Drawbacks
 - Your risk of losing money is higher
 - You need to take care of marketing
 - You need to maintain all business contracts yourself
 - Porting to multiple platforms is very difficult and time consuming
 - You may not be able to develop for consoles
 - Management and coordination require a lot of experience and expertise
 - No outside input – you may end up developing in your own bubble...

External service providers (1/4)

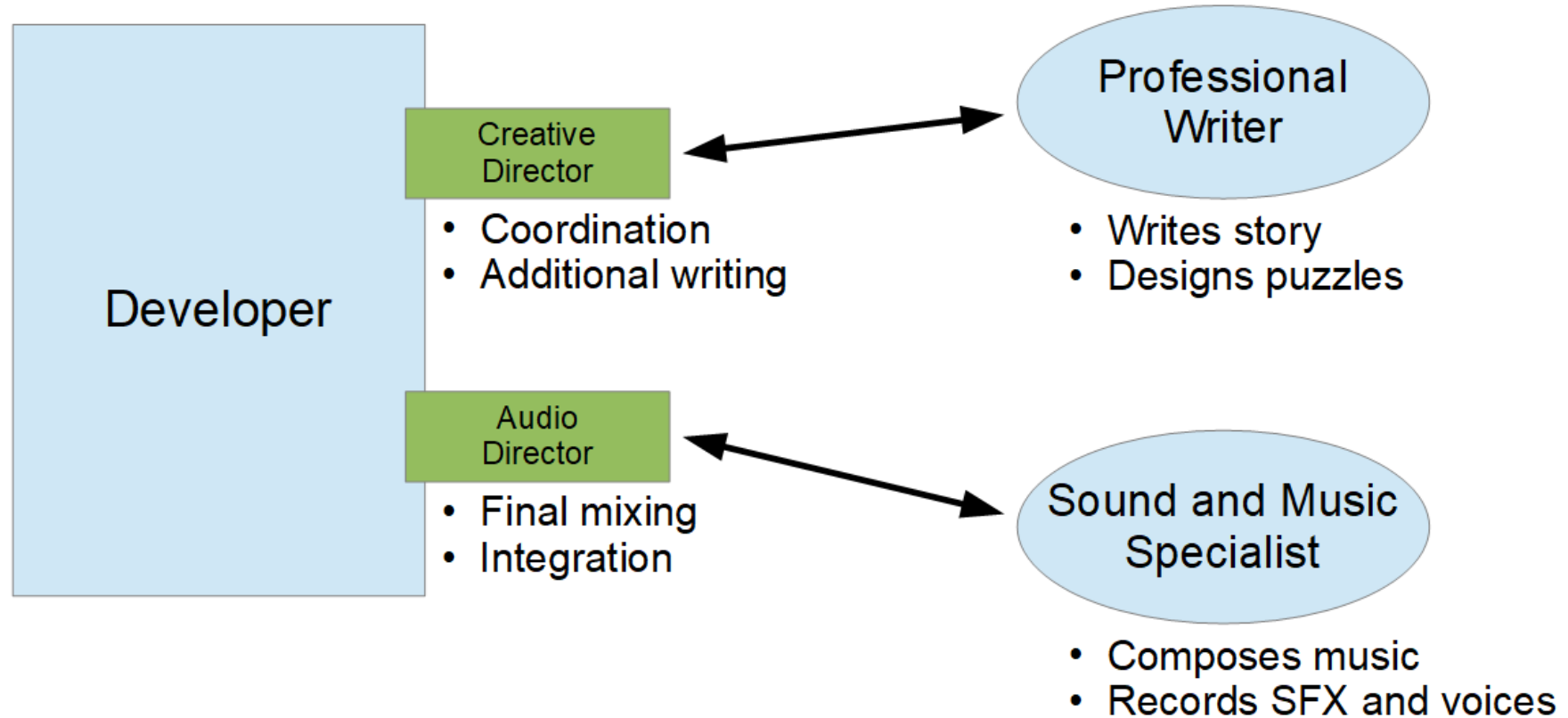


- A Service Provider (SP) is a company that provides organizations with consulting, legal, real estate, education, communications, storage, processing, and many other services.
- Developers and publishers outsource tasks to specialized SPs
 - Relief of internal resources
 - Asset creation, programming, writing, design, testing, etc.
 - Things that can't be handled internally
 - Motion capturing, voice recording, orchestration, media production etc.
 - Things that require very special skills
 - Translations, legal advice, knowledge about certain target groups, etc.

External service providers (2/4)



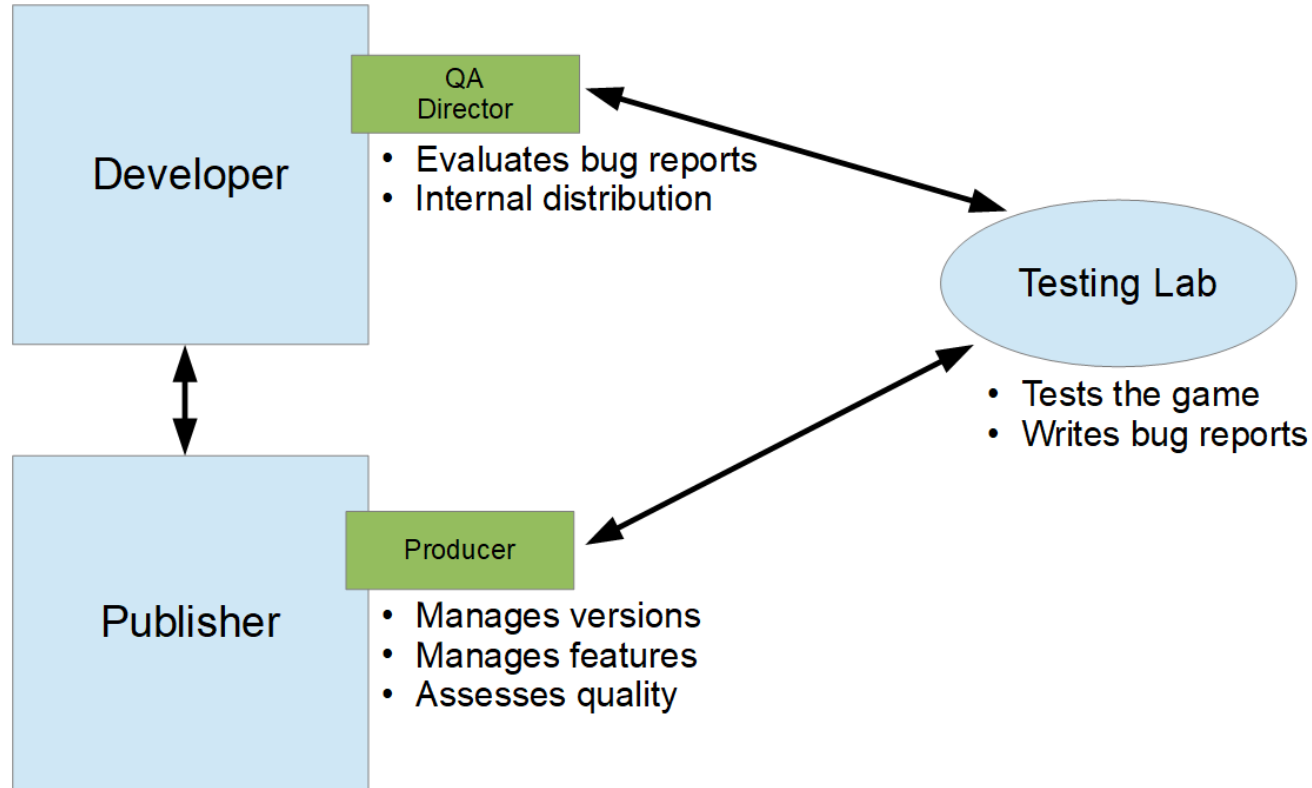
- Example 1: Developer outsources story writing and sound production.



External service providers (3/4)



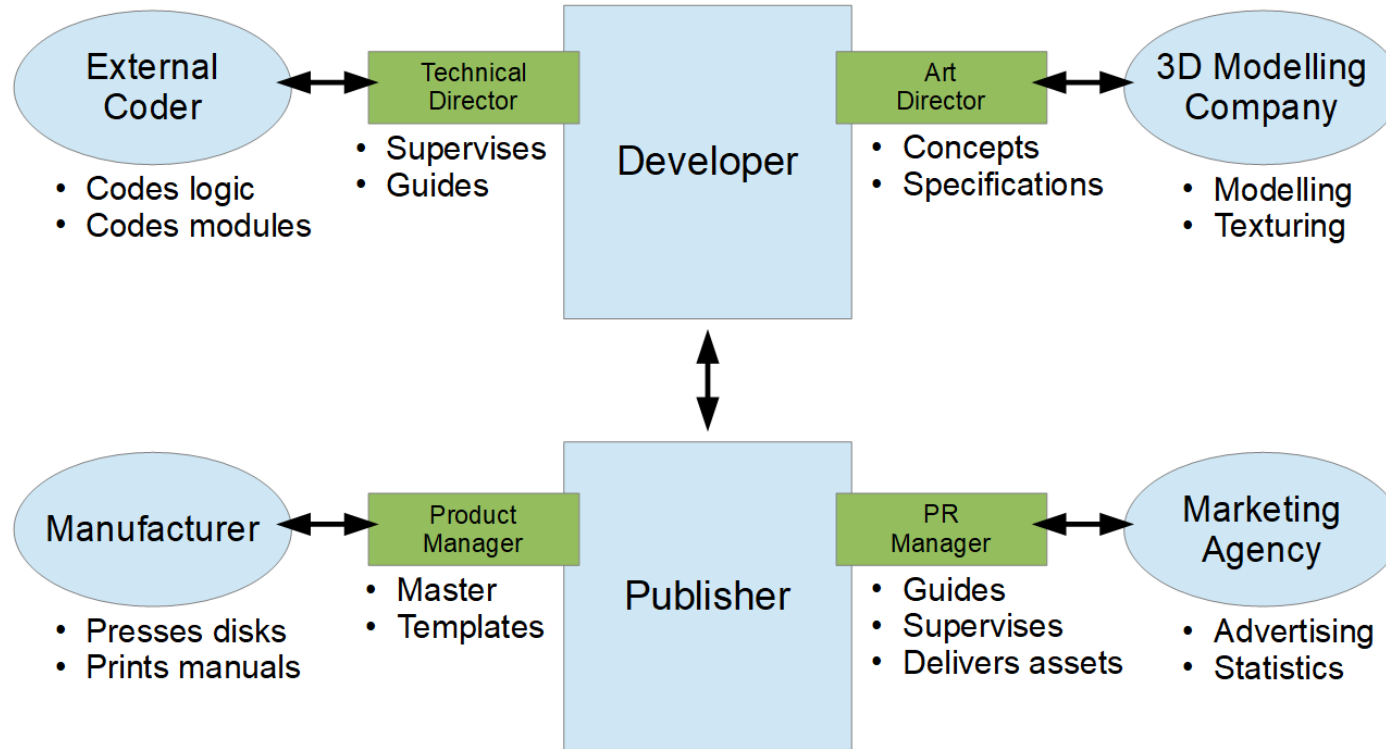
- Example 2: Developer and publisher collectively outsource QA (testing).



External service providers (4/4)



- Example 3: Developer and publisher outsource various services independently from one another.



Final thoughts...



How Game Publishers Buy Crunch Overseas



<https://www.youtube.com/watch?v=bm7KUE1Kwts>

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